

Understanding and Mitigating Text Spotting Bias in Vision-Language Models

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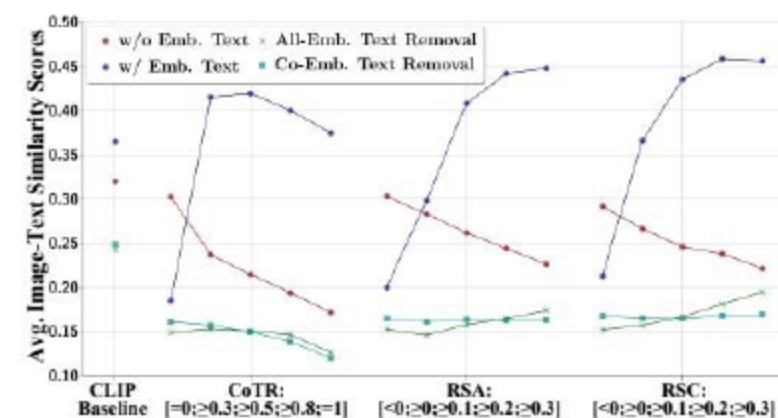
Text-Spotting Bias in CLIP

3 What Is Visual Text Bias in CLIP?



- CLIP overfits to visible text, not semantics
 - Text-cue overlap inflates similarity scores
 - LAION-2B data reflects strong text spotting
- About half of images contain visual text
 - Reliance on OCR-like cues hurts generalization
 - High scores can mask weak visual grounding

↓ Parrot Captions Amplify Text Bias



- Echoed on-image words push CLIP to text tokens.
 - Hurts vision-language representations; tasks suffer.
 - Trends vary across text-removal methods in tests.
- Reevaluate CLIP-like designs and data curation.

5 Related Work: Bias and Text-Spotting in CLIP

- Contrastive pre-training aligns image-text embeddings at scale.
- CLIP on WIT-400M/LAION is ubiquitous, yet biases persist.
- Web-scale parrot captions bias downstream performance.
 - Analyses reveal sensitivity to textual regions in CLIP.
 - Studies report color and gender biases present in CLIP.
- Efforts isolate text-spotting features shaping behavior.

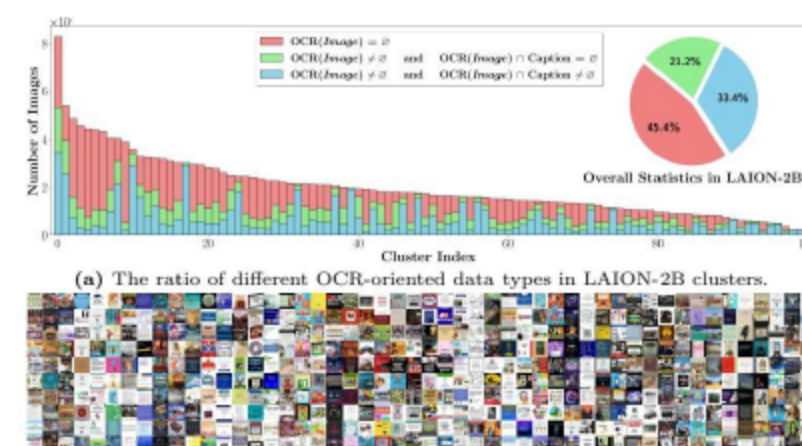
6 Key Terminology and Concepts

- Visual text bias: models over-rely on visible on-image text.
- Parrot captions: captions restate text seen in the image.
- OCR-oriented data: images where text content is central.
 - Image-text similarity: CLIP scores quantify image-caption alignment.
 - Embedded vs. co-embedded text; CoTR measures their overlap.
- Text inpainting: remove words, then compare relative scores.

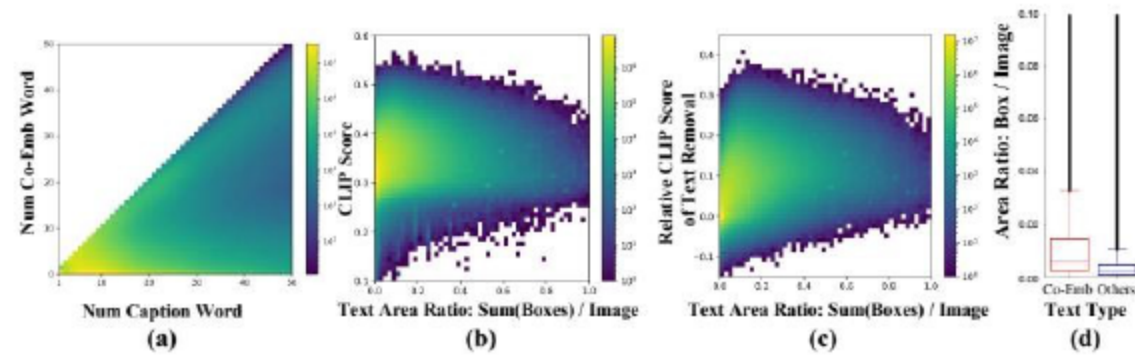
Data

Profiling LAION-2B Data

8 OCR-Oriented Content in LAION-2B



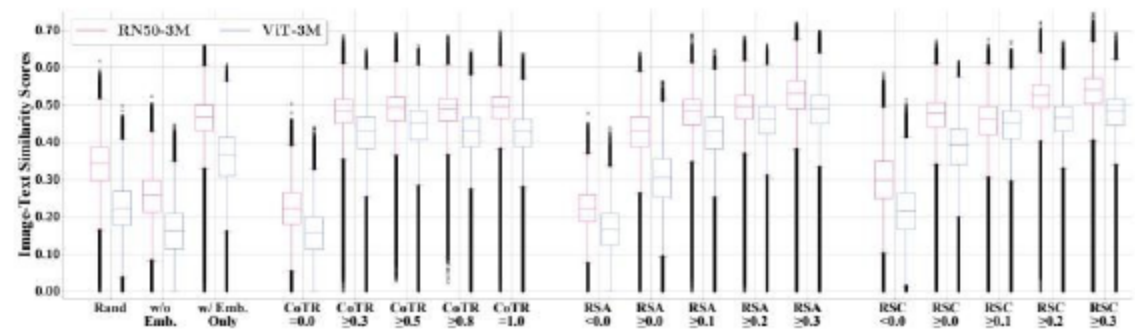
- Clusters reveal high affinity for text-dense images
 - Large fractions of clusters are text-centric content
 - Text presence strongly boosts CLIP similarity scores
- Pretraining on LAION-2B primes text-spotting focus
 - OCR-oriented types span diverse text-rich visuals
 - Distribution highlights 21.2%, 33.4%, and 45.4% shares



- Clustering LAION-2B reveals affinity for text-dense images.
 - Images containing text systematically boost CLIP scores.
 - Similarity rises with text area ratio and token overlap.
- Findings underscore CLIP's sensitivity to visible text.

Part 3

Training on Text-Curated Data



- Curated training shifts reliance from text tokens.
 - Boosts text spotting for RN50-3M and ViT-3M.
 - Hurts generalization when images lack text.
- Curation (CoTR/RSA/RSC) aids text-removed robustness.

- Web-scale data skews toward OCR-oriented content.
- Parrot captions echo text, amplifying spotting bias.
- Bias-aware curation reduces verbatim OCR text echoes.
 - Emphasize non-textual cues to strengthen grounding.
 - Use text removal and augmentation during training.
- Adopt text-aware evals; build balanced benchmarks, recipes.